MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

A Constituent Institution of Manipal University

III SEMESTER B.TECH. INTERNAL EXAMINATIONS OCTOBER 2020

MID TERM TEST

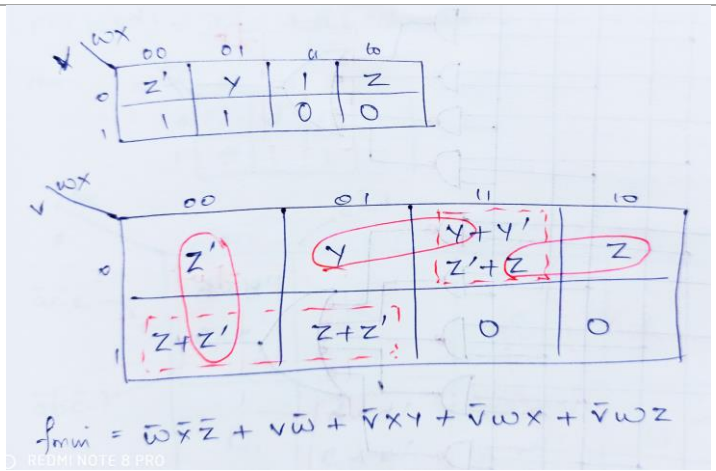
SUBJECT: DIGITAL SYSTEM DESIGN (ECE-2153)

SOLUTION

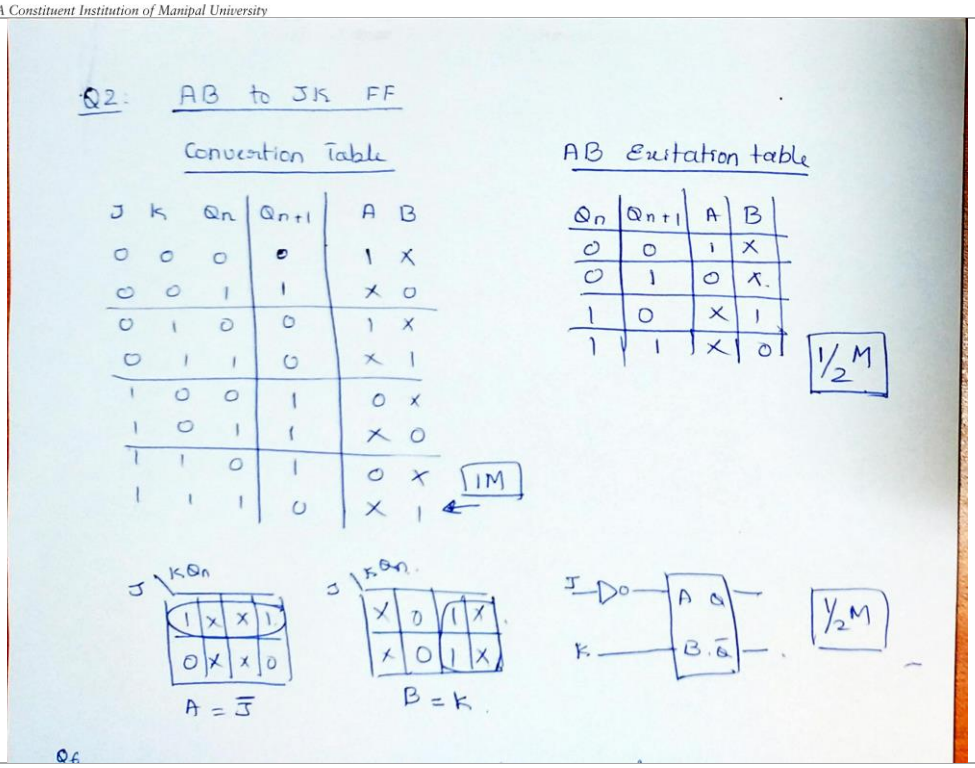
Date of Exam: **22/10/2020** Time of Exam: **8.00 AM - 9.45 AM** Max. Marks: **20**

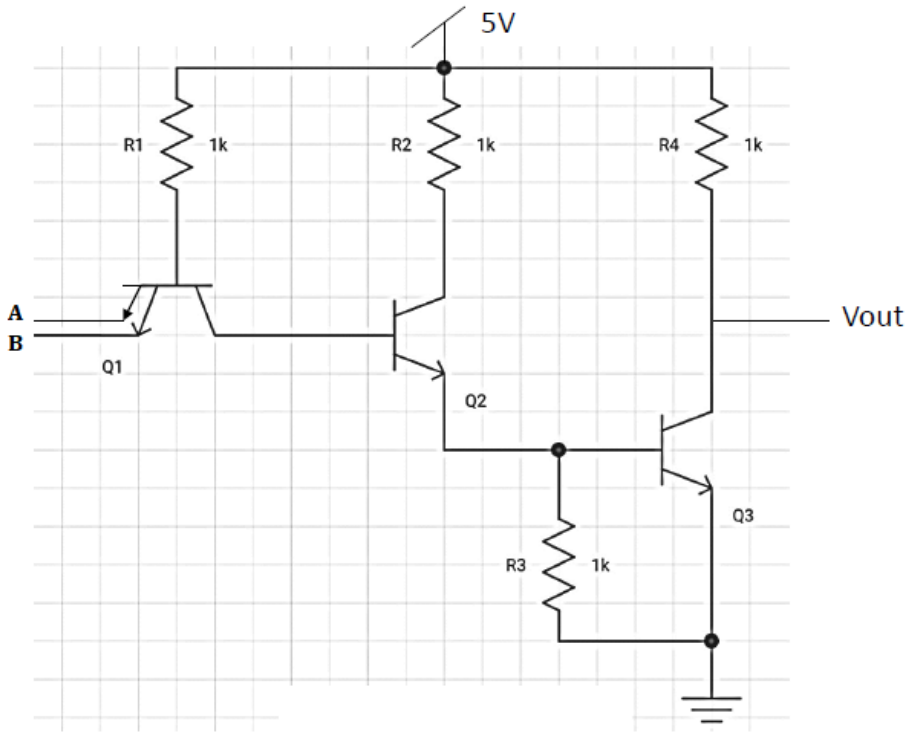
Instructions to Candidates:

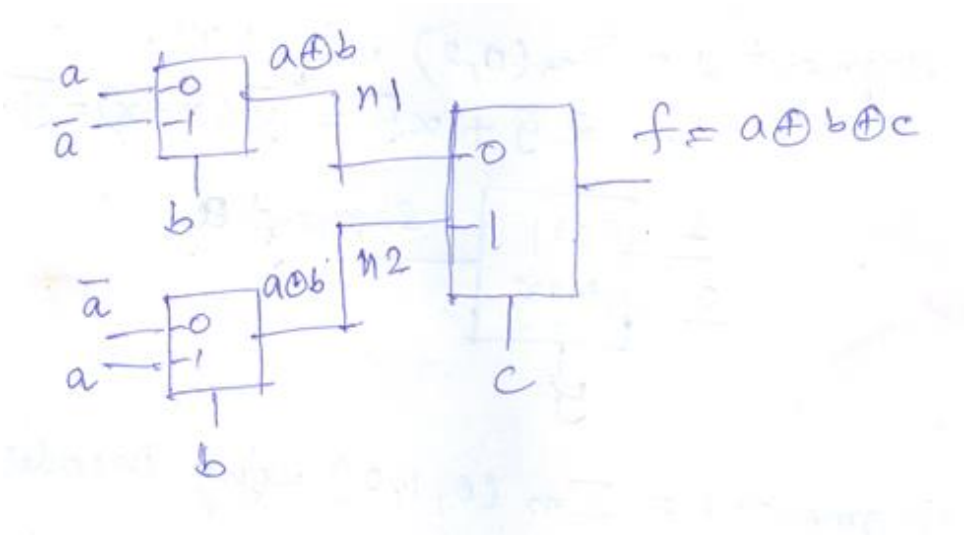
- Answer ALL the questions & missing data may be suitable assumed

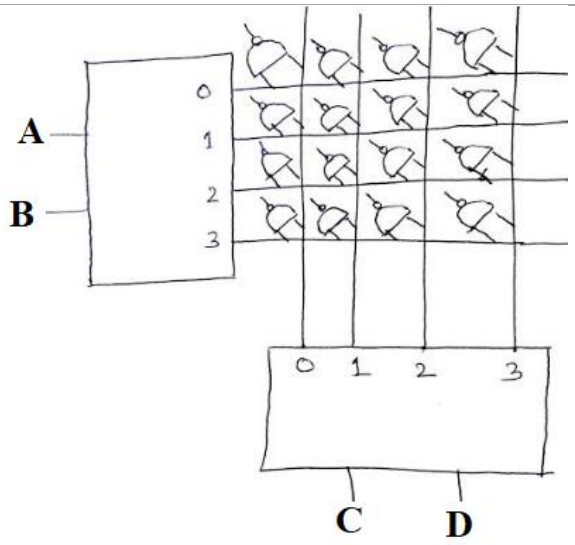
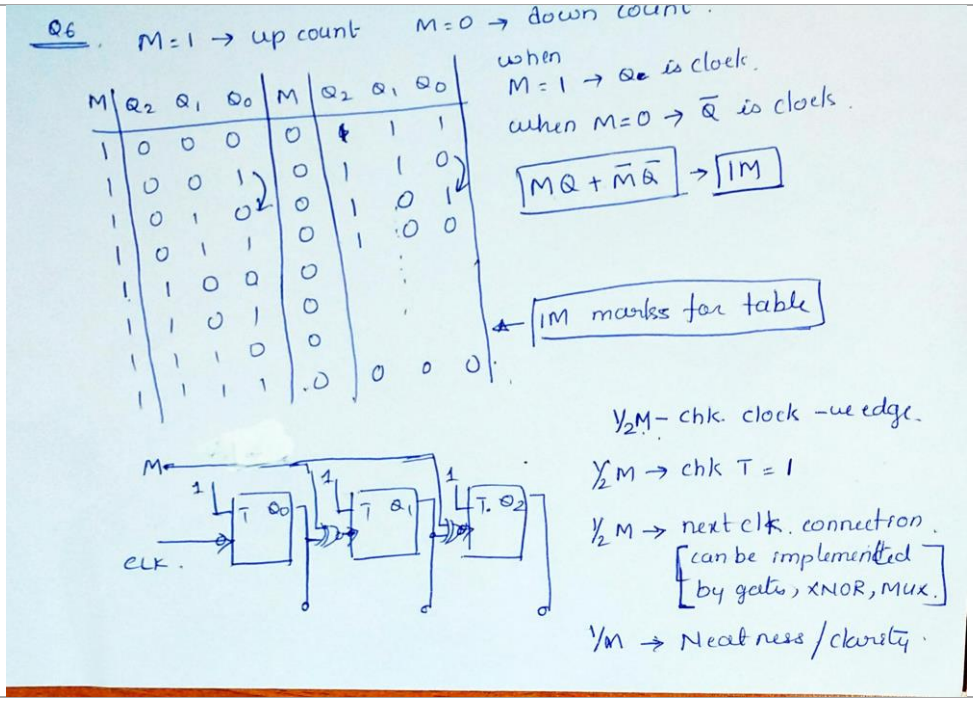
1.	Simplify the function using 3-variable VEM technique. $F(V,W,X,Y,Z) = \bar{V}\bar{W}\bar{X}\bar{Z} + \bar{V}\bar{W}XY + \bar{V}W\bar{X}Z + \bar{V}WX + V\bar{W}\bar{X} + V\bar{W}X$	**2**
A:	 **1.5M for grouping** **0.5M for equation**	
2.	Convert AB to JK flip flop. The truth table of AB flip flop is as follows **A** | **B** | **Q_{n+1}** 0 | 0 | 1 0 | 1 | Q_n' 1 | 0 | Q_n 1 | 1 | 0	**2**

A:



3.	Explain the working of a 2-input TTL NAND gate for input A=0 and B=1 with the help of a neat diagram. Why floating input in TTL logic considered as Logic "1"?	**2**
A:	2-input TTL NAND gate diagram:  Working: When inputs A=0 and B=1, transistor Q1 turns ON. Q1 pulls its collector current from base of Q2 and hence Q2 turns OFF. Since emitter does not provide the drive current to Q3, Q3 is OFF. Therefore, Vout gets logic high from the supply voltage 5V, after a small voltage drop at R4.	

	When the inputs are floating it does not help to forward bias EB junction and behaves similarly as input logic 1 applied, the EB junction of Q1 is in OFF state. **0.5M for diagram** **1M for explanation of working** **0.5M for floating input explanation**	
4.	Write a structural Verilog code for 3-bit even parity generator realized using minimum number of 2 to 1 Multiplexer ONLY. Draw the circuit diagram. Use 2 to 1 Multiplexer as component.	**3**
A:	**F= A XOR B XOR C ;** **CIRCUIT**  **(1.5)** **NOTE: OTHER CIRCUITS CAN BE CONSIDERED with minimum no. Of 2/1 MUX. IF not or other logic gates have used along with MUX, deduct ½ marks.** CODE: ***module mux_31(a,b,c,f);*** ***input a,b,c;*** ***output f;*** ***wire n1,n2;*** ***mux21 m1(a, ~a,b,n1);*** ***mux21 m2(~a, a,b,n2);*** ***mux21 m3(n1, n2,c,f);*** ***endmodule*** ***module mux21(input a,b,s,output y);*** ***wire sbar;*** ***assign sbar=~s;*** ***assign y= ((sbar & a) | (s & b));***	

	endmodule **(1.5)**	
5.	Implement active low 4 to 16 decoder with the help of two 2 to 4 (active high) decoder and sixteen 2 input NAND gate.	**3**
A:	 **3-Marks**	
6.	Design mod 8 ripple up and down counter, with control input up/down=1/0 using negative edge triggered T flip flop.	**4**
A:		

7. Consider a ALU containing two operands A and B of 4-bit. S_1S_0 are 2-bit operational code. Based on the selection of operational code, particular function will be executed. Write a behavioral Verilog code to implement the same. **4**

S_1	S_0	Function
0	0	A+B (Arithmetic)
0	1	A-B (Arithmetic)
1	0	A>>2
1	1	B<<2

A:

```
module MIDTERM_Q3(A,B,X,Y,W);
input [3:0] A,B;
input [1:0] X;
output [4:0] Y;                                    (2)
output [3:0] W;

reg [4:0] Y;

reg [3:0] W;
```

Note: Deduct ½ marks for each syntax errors.

```
always @(A,B,X)
begin

case(X)
2'd0:Y=A+B;
2'd1:W=A-B;
2'd2:W=A>>2;
2'd3:W=B<<2;                                       (2)
default:$display("invalid input");
endcase
end

endmodule
```

Note: Only behavioral modelling should be considered. IF-else statement can be also considered.